

ESP32-S3 + A7672S 4G + 2G Dev-Kit Manual

Wi-Fi + 4G LTE + 2G GSM + Bluetooth® LE IoT Communication Platform



1. Overview

The **ESP32-S3 A7672S 4G+2G LTE Development Board** is a high-performance IoT communication platform designed for advanced wireless and cellular applications. It integrates the powerful ESP32-S3 dual-core microcontroller with the reliable SIMCOM A7672S 4G LTE Cat-1 module, delivering seamless Wi-Fi, Bluetooth® Low Energy, 4G LTE, 2G GSM, and **USB Networking (RNDIS/ECM Mode)** connectivity in one compact solution

This development kit is ideal for IoT gateways, smart industrial devices, remote monitoring systems, telemetry units, cellular routers, and embedded communication products. It supports internet connectivity, **MQTT, HTTP, SMS, voice calls, RNDIS, ECM, PPP** and real-time data transfer over **LTE networks**. With dual USB ports, Nano SIM support, onboard audio interface, expansion headers, and flexible USB routing modes, this board provides complete development flexibility for engineers and product designers.

2. Key Features:

- **Multi-Connectivity Support:**
 - 4G LTE + 2G GSM cellular connectivity
 - 2.4 GHz Wi-Fi (IEEE 802.11 b/g/n)
 - Bluetooth® 5.0 (Low Energy – BLE)
- **Supported Cellular Bands:**
 - **LTE-FDD:** B1 / B3 / B5 / B8
 - **LTE-TDD:** B34 / B38 / B39 / B40 / B41
 - **2G:** GSM / GPRS / EDGE – 900 / 1800 MHz
- **Microcontroller:**
 - ESP32-S3 Dual-Core Xtensa® LX7 MCU, up to 240 MHz
- **ESP32-S3 Module:**
 - ESP32-S3-WROOM-1-N16R8
 - 16 MB Flash memory
 - 8 MB PSRAM

- **Wi-Fi Router / Gateway Function:**
 - Use ESP32-S3 as a hotspot or bridge to share 4G internet over Wi-Fi
- **Dual USB Ports:**
 - **ESP_UART** USB for programming and serial monitoring
 - **Switchable 4G_USB** port for direct **A7672S USB** access or internal USB routing between ESP32-S3 and A7672S.
- **Audio Support:**
 - Onboard **speaker and microphone pads**
 - Supports direct voice calls via A7672S without requiring external audio amplifier circuitry
- **Nano SIM Socket:**
 - Supports standard 4G **Nano SIM cards**
- **Wide Power Supply Range:**
 - Wide input voltage range: **5V–12V DC** (2A recommended)
 - Can also Power Via **USB**
- **UART Level Translation:**
 - Safe UART communication between ESP32-S3 and A7672S
- **Onboard Status Indicators:**
 - **GPIO12:** 4G Network Status LED
 - **GPIO11:** Wi-Fi Status LED
 - **GPIO14:** System Status LED
 - **NETLIGHT LED:** Indicates A7672S network activity
 - **POWER LED:** Indicate the PCB Input Supply
- **44-Pin Expansion Header (2 × 22, 2.54 mm pitch):**
 - Provides access to all ESP32-S3 GPIOs
 - Includes GPIO connections to the 4G module via removable 0 Ω link resistors for flexible routing.
- **4G debug Pads:**
 - Onboard **4G_debug** PADS to debug or monitor the A7672S status

- **Communication Interfaces:**

- UART, USB, Wi-Fi, Bluetooth LE, 4G LTE, and GSM
- Supports standard ESP32-S3 peripheral interfaces (GPIO, SPI, I²C, I²S, PWM, ADC)
- USB Networking Support (**RNDIS / ECM / PPP Mode**)

- **Supports Windows/Linux Dial-up Internet**

- Integrated USB networking support for Windows and Linux systems, enabling dial-up internet connectivity via the A7672S module.

3. Board Hardware Overview

Component	Description
ESP32-S3 MCU	Dual-core Wi-Fi + BLE microcontroller
SIMCOM A7672S 4G Module	LTE Cat-1 module with optional GNSS
Voltage Translation	UART level translator between ESP32-S3 and A7672S
USB-to-UART Bridge	USB-to-UART bridge for ESP32-S3 programming
USB_SELECT Switch	Selects operating mode of the 4G_USB port
Nano SIM Socket	Nano SIM card interface for cellular connectivity
Speaker & Microphone Pads	Onboard audio interface for voice call support
USB Ports (x2)	USB1 for ESP32-S3, USB2 switchable for A7672S or ESP32 USB routing to 4G USB

3 Status LEDs	Wi-Fi, 4G, and System activity indication (When works in Mode - 3)
1 Network LED	Indicates network activity and registration status of A7672S
1 Power LED	Indicates presence of input power supply
RESET Switch	Resets the ESP32-S3 microcontroller
Boot Switch	Puts ESP32-S3 into bootloader (download) mode
UFL male Connectors(x2)	Dedicated connectors for LTE antenna and GNSS antenna

4. GPIO Mapping

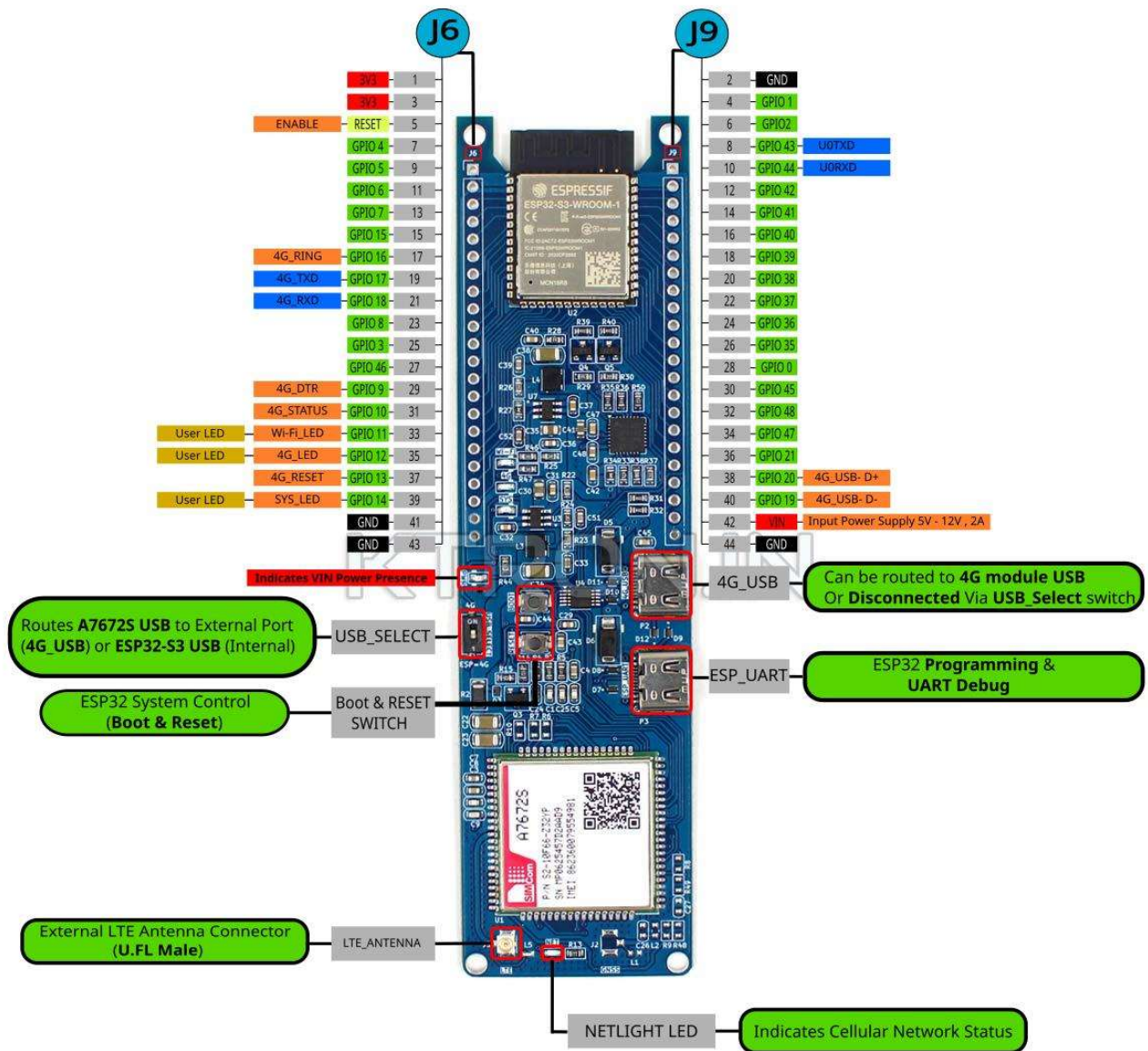
ESP32-S3 GPIO	Direction	Connected To (A7672S Pin)	Signal Name	Description
GPIO17	RX	Pin 9	TX	ESP32 receives data from 4G module
GPIO18	TX	Pin 10	RX	ESP32 transmits data to 4G module
GPIO16	Input	Pin 4	RI (Ring Indicator)	Indicates incoming call/SMS
GPIO19	USB D-	Pin 28	USB D-	USB data line (negative)
GPIO20	USB D+	Pin 27	USB D+	USB data line (positive)
GPIO9	Output	Pin 3	DTR	Data Terminal Ready control
GPIO10	Input	Pin 66	STATUS	Indicates module operational status
GPIO13	I/O	Pin 16	4G_RST	Reset Pin of the A7672S
GPIO11	Output	—	Wi-Fi LED	Indicates Wi-Fi status
GPIO12	Output	—	4G LED	Indicates LTE status
GPIO14	Output	—	System LED	Indicates power/system activity

—	RX	Pin 22	4G_Debug1 RX	Receives debug commands or control data for internal diagnostics
—	TX	Pin 23	4G_Debug1 TX	Outputs internal debug logs, trace messages, and diagnostic information
—	INPUT	Pin 42	MIC_P	Positive (+) Pin for Microphone input
—	INPUT	Pin 43	MIC_N	Negative (-) Pin for Microphone Input
—	Output	Pin 40	EAR_P	Speaker output positive (+)
—	Output	Pin 41	EAR_N	Speaker output negative (-)

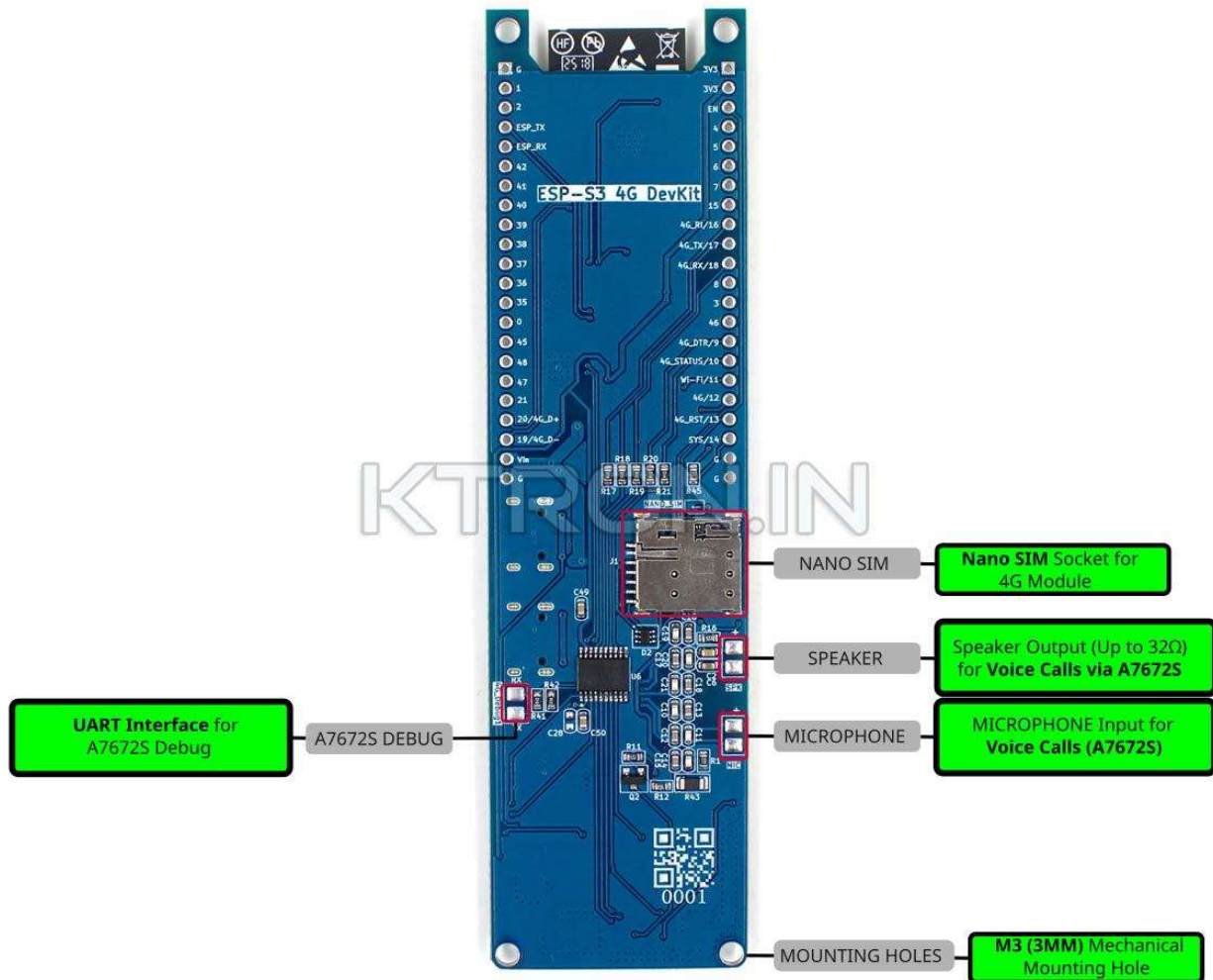
Note- All other remaining GPIO's Are free to use

5. Pin Out Details

Front Side



Back Side



6. USB Port Functions

- **ESP_UART** (ESP32-S3): Programming and Serial Monitoring
- **4G_USB** (Switchable via **USB_Select** Switch):
 - Mode 1: Direct A7672S USB or
 - Mode 2: Externally Disconnected (Internally Connected with ESP32-S3 USB to 4G USB)

7. Audio Interface

- Speaker: up to 32Ω
- Microphone: Electret condenser MIC
- Supports voice calls via A7672S without extra circuitry

8. USB_Select Switch

- Controls 4G_USB routing.
 - **4G mode:** Direct AT command access to A7672S
 - **ESP-4G mode:** Internal ESP32-4G USB connection, no direct user access

Working Modes of the Development Board

Mode 1: A7672S with ESP32-S3 (Access 4G Using ESP32-S3)

- **Description:**

In this mode, the **ESP32-S3** is connected **serially (UART)** with the **A7672S 4G module**. You can **load custom firmware or code onto the ESP32-S3** to make the development board function exactly as you want—whether as an **IoT device**, a **cellular-enabled sensor**, or even a **basic 4G communication device**.

- **Capabilities:**

- **Serial communication with A7672S:** Full control over the 4G module via UART using AT commands.
- **Custom programming:** Run any code on ESP32-S3 for IoT applications, data collection, remote monitoring, or control tasks.
- **Supports multiple protocols:** HTTP, MQTT, SMS, CALL and more.
- **Flexible device functionality:** Can be programmed as a smart sensor, cellular device, or any embedded system requiring 4G connectivity.

- **Example:**

```
#include <HardwareSerial.h>
HardwareSerial mySerial(2);

#define RESET_PIN 13 // Optional reset control (connect if needed)

bool moduleReady = false;

void setup() {
  Serial.begin(115200);
  delay(500);

  // Start UART2 for A7672S module
  mySerial.begin(115200, SERIAL_8N1, 17, 18);
  delay(500);
```

```

pinMode(RESET_PIN, OUTPUT);
digitalWrite(RESET_PIN, HIGH);

Serial.println("=== A7672S AT Command Test ===");
Serial.println("Waiting for *ATREADY: 1 ...");

waitForModuleReady();

Serial.println("☑ ☑ ☑ ☑ Module ready. You can now type AT commands: ☑☑☑☑");
Serial.println("-----");
sendATCommand("AT");
}

void loop() {
  // Allow sending commands only when module is ready
  if (moduleReady && Serial.available()) {
    String cmd = Serial.readStringUntil('\n');
    cmd.trim();
    if (cmd.length() > 0) {
      sendATCommand(cmd);
    }
  }

  // Always read responses from module
  if (mySerial.available()) {
    String response = mySerial.readStringUntil('\n');
    response.trim();
    if (response.length() > 0) {
      Serial.print("<< ");
      Serial.println(response);
      // Detect *ATREADY: 1 if it comes later
      if (response.indexOf(" *ATREADY: 1 ") != -1 && !moduleReady) {
        moduleReady = true;
        Serial.println("\n☑ Module is ready for AT commands! ☑");
        Serial.println("-----");
      }
    }
  }
}

// --- Wait for *ATREADY: 1 before proceeding ---
void waitForModuleReady() {
  unsigned long start = millis();
  const unsigned long timeout = 30000; // 30 seconds

  while (millis() - start < timeout) {

```

```

if (mySerial.available()) {
    String response = mySerial.readStringUntil('\n');
    response.trim();
    if (response.length() > 0) {
        Serial.print("<< ");
        Serial.println(response);
        if (response.indexOf("*ATREADY: 1") != -1) {
            moduleReady = true;
            return;
        }
    }
}

Serial.println(" ⚠ Timeout waiting for *ATREADY: 1");
moduleReady = false;
}

// --- Helper function to send AT command ---
void sendATCommand(String cmd) {
    Serial.print(">> ");
    Serial.println(cmd);
    mySerial.println(cmd);
}

```

Mode 2: A7672S Standalone (Direct 4G Module Access)

- **Description:** Provides direct access to the A7672S 4G module for AT command communication, testing, configuration, firmware updates, and network diagnostics without ESP32 intervention.
- **Setup:**
 - Set the **USB_Select** switch to **4G**.
 - Connect the USB cable to the **4G_USB** socket.
 - Set serial monitor baud rate to **115200**
- **Capabilities:**
 - Send and receive **AT commands** directly to the A7672S module.
 - Update **A7672S firmware**.
- **Example:**
AT

OK

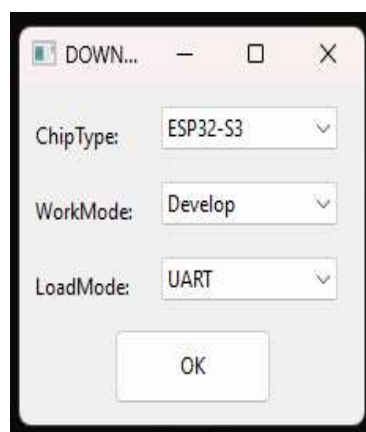
AT+CSQ

+CSQ: 30,99

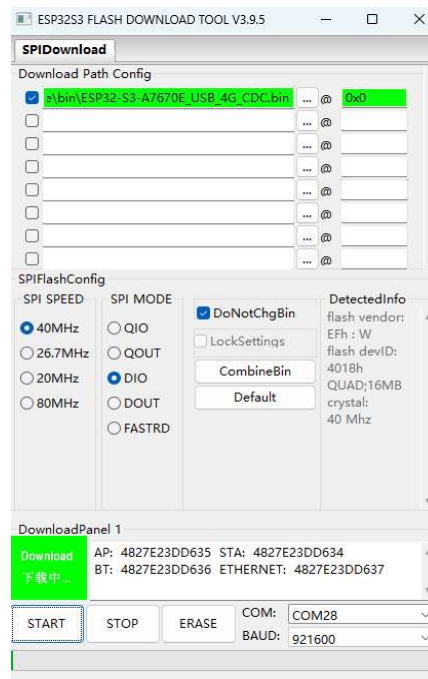
OK

Mode 3: Wi-Fi Router Mode

- **Description:** Use the development board as a **Wi-Fi hotspot**, allowing other devices to access the internet through the **A7672S 4G module**.
- **Setup:**
 - Set the **USB_SELECT** switch to the **ESP-4G** position.
 - Insert a **Nano SIM card** with a valid data plan into the SIM holder.
 - Connect the USB cable to the **ESP_UART** USB port (ESP32 programming port).
 - Open the [Flash Download Tool](#) on your PC.
 - For firmware access, please contact our sales at sales@ktron.in.
- **Capabilities:**
 - Creates a local Wi-Fi network (Access Point mode).
 - Routes internet traffic through the A7672S 4G module.
 - Allows multiple devices to connect and access the internet.
- **Example:**
 - Now, after completing the above setup, open the **Flash Download Tool**. You will see a window like the image below. Keep all the options unchanged and press **OK**



- In the next window, select the appropriate firmware file and set the flash address to **0x0**.



- Next, choose the correct **COM port** corresponding to the connected development board. Leave all other parameters unchanged and click **Start** to begin the flashing process. After a few seconds, the tool will display **FINISH**, indicating that the firmware has been successfully programmed.
- Once flashing is complete, press the **RESET** button on the board.
- Upon restart:
 - The **SYS** and **4G** onboard LEDs will begin blinking.
 - After network initialization, the **NETLIGHT** LED will also start blinking, indicating successful cellular network registration.
- You may open the **Serial Monitor** to observe continuous successful network ping messages, as illustrated in the image below.
- The board will now operate as a Wi-Fi hotspot

- Open the **Serial Monitor** to verify network connectivity.
 - During initialization, you may observe temporary ping timeouts such as shown in image This behaviour is normal during network registration.
 - Once the 4G connection is fully established, successful ping responses will appear show in image below.

```
I (15448) 4g_main: rssi=31, ber=99
I (15448) 4g_main: changed: ap dns addr (auto): 49.45.0.1
I (15452) 4g_main: Ping addr 8.8.8.8 Restart..
W (17456) 4g_main: From 8.8.8.8 icmp_seq=1 timeout
W (17456) 4g_main: Consecutive ping timeouts: 1/5
I (25554) 4g_main: rssi=30, ber=99
I (25554) 4g_main: Ping addr 8.8.8.8 Restart..
W (27554) 4g_main: From 8.8.8.8 icmp_seq=1 timeout
W (27554) 4g_main: Consecutive ping timeouts: 2/5
I (35655) 4g_main: rssi=29, ber=99
I (35655) 4g_main: Ping addr 8.8.8.8 Restart..
I (35706) 4g_main: 64 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=50 ms

I (45706) 4g_main: rssi=29, ber=99
I (45706) 4g_main: Ping addr 8.8.8.8 Restart..
I (45742) 4g_main: 64 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=35 ms

I (55743) 4g_main: rssi=29, ber=99
I (55743) 4g_main: Ping addr 8.8.8.8 Restart..
I (55781) 4g_main: 64 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=37 ms

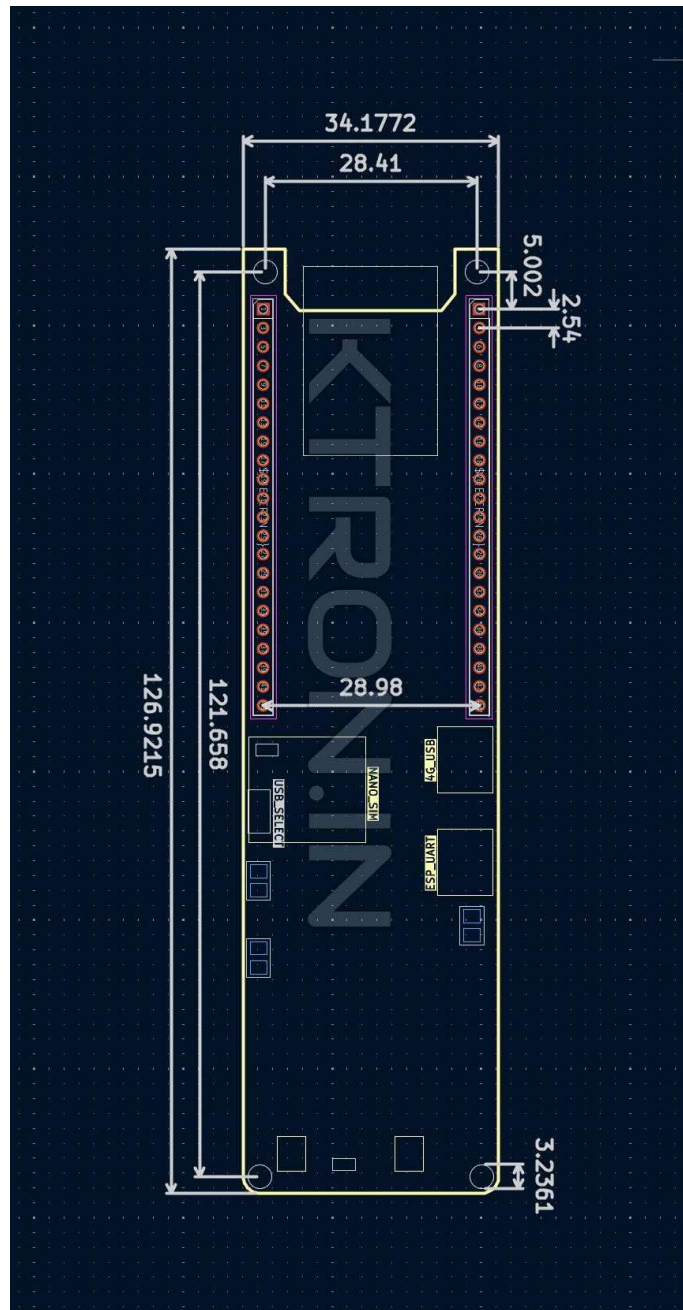
I (65782) 4g_main: rssi=29, ber=99
I (65782) 4g_main: Ping addr 8.8.8.8 Restart..
I (65827) 4g_main: 64 bytes from 8.8.8.8 icmp_seq=1 ttl=113 time=44 ms

I (75828) 4g_main: rssi=29, ber=99
I (75828) 4g_main: Ping addr 8.8.8.8 Restart..
I (75875) 4g_main: 64 bytes from 8.8.8.8 icmp_seq=1 ttl=112 time=46 ms
```

- **Wi-Fi Access**
 - The board will now operate as a Wi-Fi hotspot with the following credentials:
 - **SSID:** esp_4g_router
 - **Password:** 12345678

Connect a laptop, smartphone, or other Wi-Fi-enabled device to this network to access the internet via the 4G connection.

Unit: mm



Product Identification: ESP32-S3 + A7672S 4G + 2G Dev-Kit

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